

Fuzzy Logic PID Controller for Missile Terminal Guidance

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Abstract - We demonstrate fuzzy logic terminal guidance for a surface-to-surface missile. A conventional proportional-integral-derivative (PID) guidance scheme is first employed to form the basis for the fuzzy terminal guidance. The PID guidance scheme employs heading and flight path angle errors derived from sensor imagery of target location to compute angular rate commands. The fuzzy logic terminal guidance design procedure involves translating the conventional PID guidance scheme into a fuzzy control structure with the associated fuzzification of input/output variables, encoding of decision-making knowledge into rulebases, and defuzzification. A novel scheme is implemented in which fuzzy rulebases are used to generate and combine PD and PI portions of the guidance commands. Additionally, the fuzzy terminal guidance includes rulebases to control speed and to perform gain scheduling. Performance evaluations demonstrate the feasibility of the approach.

I. INTRODUCTION

In recent years, rule-based systems have become increasingly popular as practical applications of artificial intelligence. These expert systems have performed as well as humans in several problem domains ([1], [2], and [3]). However, their lack of flexibility in representing the subjective nature of human decision-making limits their performance in domains where it is difficult to formulate crisp rules, and situations for acting on those rules. This uncertainty, inherent in human decision-making, can be incorporated into expert systems via fuzzy set theory [4]. In fuzzy set theory, linguistic variables can represent abstract or subjective concepts. Linguistic variables, and thus the uncertainty associated with human decision-making, have been incorporated into expert systems in the form of fuzzy logic reasoning.

Fuzzy reasoning are rule-based systems that use fuzzy linguistic variables to model a human's rule-of-thumb approach to problem solving. Fuzzy reasoning includes rules to direct the decision process and membership functions to convert linguistic variables like *high* and *low* into precise numeric values a computer needs. The rule set is gleaned from a human expert's knowledge, which is generally based on his experience. The developer chooses the membership functions to represent the

human expert's interpretation of the linguistic variables.

The seminal work of [5] provided for the first real application of fuzzy control. Since then numerous applications of fuzzy logic and fuzzy control have appeared. In [6] a fuzzy controller was developed for an automatic train operation system. Results showed the fuzzy controller outperforming conventional PID controllers. As a matter of fact, their controller was implemented in the Sendai railway system in 1987 and has proved very successful. In [7] a scheme for weighting PI and PD controllers is introduced. Gain scheduling and pre-compensation of conventional PID controllers using fuzzy logic is demonstrated in [8] and [9], respectively. Aerospace applications of fuzzy control include an aircraft autopilot controller [10], a fuzzy guidance scheme for an automatic carrier landing system [11], and fuzzy gain scheduler for flight control [12].

In this paper we present a fuzzy controller for surface-to-surface missile terminal guidance. The problem domain involves guiding a low-altitude missile to impact a ground based target (e.g., tank) using information from an inertially mounted imaging sensor. We first begin with a discussion of the design procedure for developing the fuzzy terminal guidance, followed by a description of the design itself. Performance evaluations of the fuzzy and conventional guidance schemes are then demonstrated for several scenarios including stationary and non-stationary target.

II. FUZZY GUIDANCE DESIGN PROCEDURE

Figure 1 presents an overview block diagram of the design procedure used to develop the fuzzy terminal guidance system. The procedure begins with specification of platform dynamics of the missile and target models. Next comes specification of the imaging sensor which generates the requisite measurements of target/platform states for guidance. Based on target information from the imaging sensor, an off-line guidance design is then required. In this case, we implement a Proportional-Integral-Derivative (PID) guidance scheme. After the guidance design, the design procedure then proceeds along parallel paths: the upper path of figure 1 is for the conventional guidance approach, while the lower is for the fuzzy-based approach. In the following sections, we describe in further detail the modules of figure 1. Complete details of the design proce-

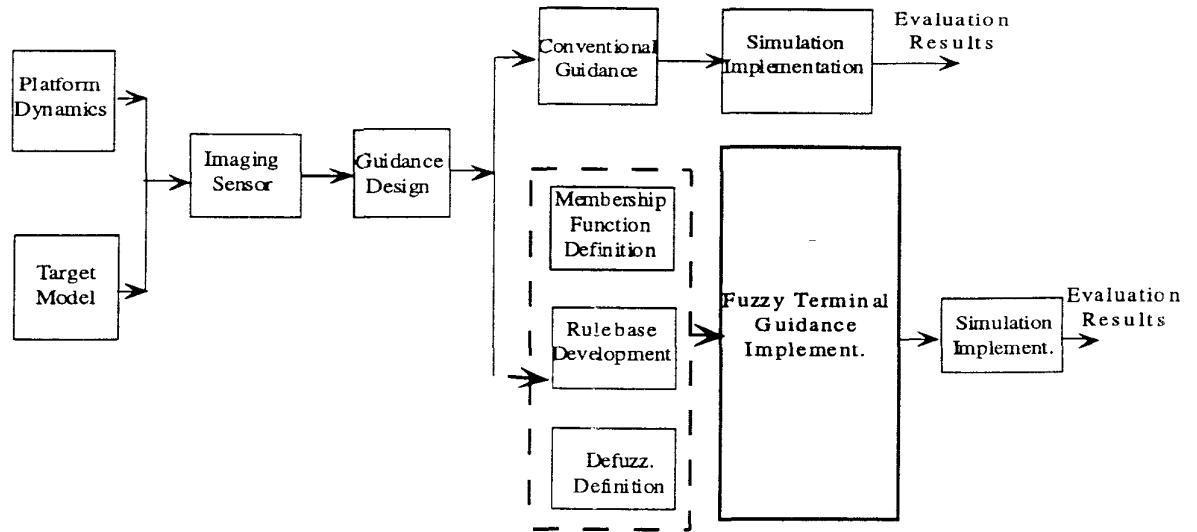


Fig. 1: Fuzzy Terminal Guidance Design

ture and module description may be found in [13].

A. Platform Dynamics

The platform dynamics module supports the simulation of the missile platform. The module implements a simplified six degree-of-freedom (6 DOF) model of missile response to guidance system speed, pitch, and bank commands, and generates platform position and velocity in 3D-space. Submodules within the platform module include: 1) a rotational dynamics block which converts missile body rates into the Euler Angle set comprised of roll, pitch, and yaw; 2) a translational dynamics block which combines the aerodynamic parameters, the thrust level, and the Euler angles to generate the basic vehicle velocity parameter vector comprised of speed, flight path angle, and heading; 3) a translational kinematics block which generates the vehicle position and velocity vectors in the local navigation frame; and 4) the aero-parameter module, which uses the vehicle states to generate lift and drag, as well as angle-of-attack and sideslip. Additional submodules include guidance and thrust controller modules which we discussed later.

B. Target Model

The target model module supports the simulation of a target and generates a time history of target position, velocity, and acceleration in 3D-space. The module supports definition of a forcing function for target motion.

C. Imaging Sensor

For the imaging sensor, we limit our attention to a passive sensor like a TV seeker and assume that image pre-processing occurs such that a discrete set of target points are projected on an image screen. These points

refer to the edges of the target, which we model as a right rectangular prism (i.e., a 3D shape with all face rectangular). We further assume that the sensor is inertially mounted along the longitudinal axis of the missile and has a field-of-view of ± 90 degrees in both lateral and vertical axes.

The processing data pathway for the imaging sensor begins first by calculating the relative position vector between the target and the platform in the body frame for all target points. The relative vector is then, in turn transformed into the navigational and then into the sensor frame. Next, the 3D relative position vector, is projected onto the 2D image plane, which is at a focal length, f , from the sensor coordinate origin. The geometry associated with this projection is shown in figure 2. From geometry, and using similar triangles, the position vector on the image plane, (u,v) , for each point is given by:

$$\begin{aligned} u &= f \frac{z_s}{x_s} \\ v &= f \frac{z_s}{x_s} \end{aligned} \quad (1)$$

where (x_s, y_s, z_s) is the 3D relative position vector in the sensor frame.

III. FUZZY TERMINAL GUIDANCE DESIGN

We now detail the off-line design of the fuzzy terminal guidance system. We begin by specifying the conventional guidance and thrust controllers and then detail the specification of the fuzzy guidance and speed controllers, including membership function definition, rule-base development, and defuzzification definition.

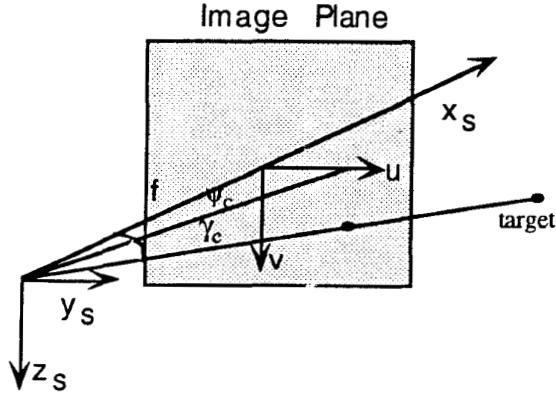


Fig. 2: Geometry for Projection onto Image Plane

A. Conventional Guidance Design

Guidance Design:

The guidance module of the platform dynamics controls the missile body rates (ω) to achieve a desired heading (ψ_c) and flight path angle (γ_c), as shown in figure 2, and to null sideslip (β) angle. The desired heading and flight path angles are calculated based on image sensor measurements, i.e., based on the (u, v) measurement pairs. To generate the desired ψ_c and γ_c , we use the average or mean values of u and v 's for all target points and then calculate the angle deviations of this mean target position from the image center. That is, we assume since the sensor is mounted on the longitudinal axis of the missile, keeping the target mean point on the center of the image plane will result in target impact. Therefore, the (ψ_c, γ_c) pair is calculated as:

$$\begin{aligned}\psi_c &= \tan^{-1} \left(\frac{\bar{u}}{\bar{f}} \right) \\ \gamma_c &= \tan^{-1} \left(\frac{-\bar{v}}{\bar{f}} \right)\end{aligned}\quad (2)$$

where $\bar{\cdot}$ denotes mean values.

Given the desired heading and flight path angle, we implement a PID guidance system, i.e:

$$\omega = \begin{bmatrix} p \\ q \\ r \end{bmatrix}^T = \begin{bmatrix} k_p^{\psi} \cdot \psi_e + k_i^{\psi} \cdot \int \psi_e dt + k_d^{\psi} \cdot \dot{\psi}_e \\ k_p^{\gamma} \cdot \gamma_e + k_i^{\gamma} \cdot \int \gamma_e dt + k_d^{\gamma} \cdot \dot{\gamma}_e \\ k_r \cdot \beta \end{bmatrix} \quad (3)$$

where the e subscript refers to error (i.e., desired minus actual), ω is the vector of the commanded missile body rates, and the k 's are gains to chosen to provide good response for the target/platform geometries considered.

Speed Control:

The thrust controller module generates a thrust command to maintain a constant speed command V_c . For the conventional guidance, constant values of V_c within a range from 300 ft/s to 500 ft/s were specified. For the fuzzy terminal guidance, we either employed a constant speed command, as in the conventional guidance, or V_c was varied using a fuzzy guidance speed control. Specifics of the fuzzy guidance speed controller will be discussed in the section that follows.

B. Fuzzy Guidance Design

Guidance Design:

Figure 3 shows the architecture employed for the fuzzy terminal guidance system. The architecture is duplicated for both the heading (ψ) and flight path angle (γ) channels. As shown, guidance path errors drive, in parallel, a Proportional-Derivative (PD) controller and a Proportional-Integral (PI) controller. The results of the PD/PI controllers (u_{PD} and u_{PI} , respectively) are combined via a Fuzzy Weighting rulebase. The method for combining in parallel PD and PI controllers is due to [7]. However, whereas they employed a simple switching logic, we use a fuzzy rulebase to combine controller outputs. The combined control (u_{total}) is then processed via a gain scheduler to account for variations over the flight envelope.

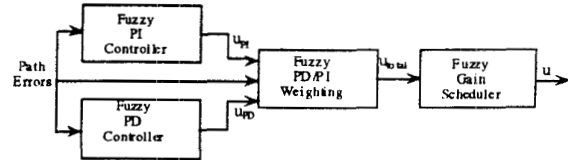


Fig. 3: Fuzzy Terminal Guidance System

Fuzzy Speed Control:

Figure 4 details the processing pathway for the fuzzy speed controller. As shown, range to target and heading error (ψ_e) are used to determine the speed setting (V_c). The allowable speed range is from 300 to 500 ft/s.

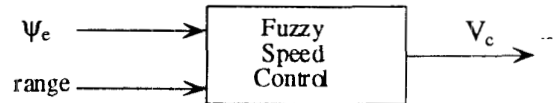


Fig.4: Fuzzy Speed Controller

C. Fuzzy Terminal Guidance Implementation

The lower pathway of figure 1 represents the development and definition of the fuzzy logic gain scheduler. This is composed of three sub-blocks: 1) the membership function definition; 2) the fuzzy rulebase development; and 3) the defuzzification definition. We briefly

describe these sub-blocks below.

The membership function definition performs the mapping from the deterministic or crisp numerical data into fuzzy sets. Fuzzy sets are created for the six PD/PI controller inputs (error, error rate, and integral error for both the ψ and γ channels), the four PD/PI outputs (corresponding to the two commanded body rates, p and q from (3) for both PD and PI portions), the range to target input, the fuzzy PD/PI weighting factor, the fuzzy gain scheduler, and the fuzzy speed control setting, V_c . Seven fuzzy sets are used for all parameters, except the weighting factor and gain scheduler which use four fuzzy sets, and range and speed which use three fuzzy sets. The seven fuzzy sets correspond to the linguistic variables Negative Big (NB), Negative (N), Negative Small (NS), Zero (Z), Positive Small (PS), Positive (P), and Positive Big (PB). The four fuzzy sets for the weighting factors and gain schedulers correspond to Low, Low Medium, High Medium, and High, where Low for weighting factor corresponds to total PI control and High to total PD control. The three fuzzy sets for range correspond to Near Range, Mid Range, and Far Range and the fuzzy sets for V_c are Low, Medium, and High. A combination of Gaussian, sigmoidal, and trapezoidal membership functions is employed for all fuzzy sets. Figure 5 shows, as an example, the fuzzy sets for the heading error rate.

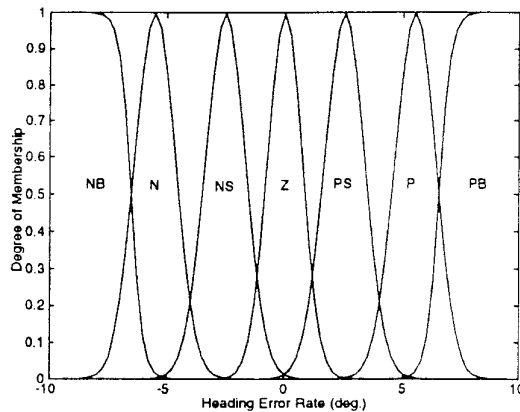


Fig. 5: Fuzzy Speed Controller

The rulebase development sub-block of figure 1 defines the fuzzy decision-making logic. Each rule is an IF-THEN structure, where the IF part is called the antecedent and the THEN part is called the consequent. For the PD/PI controllers and the weighting rulebases the error and error rate, error and integral error, and error and error rate are the antecedents, respectively, and the guidance commands are the consequents for the controllers. A weighting factor, is the consequence for the weighting rulebase. Since there are two antecedents and

one consequent per rule, and since each antecedent has partial membership in seven fuzzy sets, there are 49 fuzzy rules for each of these rulebase associated with each guidance channel (i.e., heading and flight path angle). In a similar fashion, the gain scheduling rulebases and the speed control rulebase have 21 rules each. We use Mamdani's max-min mechanism to perform the inferencing.

As an example, figure 6 shows how the two weighting PD/PI rulebases are structured with the two inputs (error and error rate) and the one output (weighting factor). The rulebase is shown in a matrix form with the seven fuzzy sets for error in the first column (left of the dark-line) and the seven fuzzy sets for error rate in the bottom row (below the dark line). Rulebase definition involves filling in each of these 49 matrix elements with the corresponding fuzzy set for the weighting factor. This is done by applying heuristic knowledge that if error and error rate are large, then weigh the PD output more (i.e., weighting factor is High). If we are at a steady state condition (i.e., error and error rate are small), weigh the PD output less (i.e. weighting factor is Low), and thus weigh the PI output to a greater extent.

Positive Big	High	High	High	High	High	High	High
Positive	High	High Medium	Low Medium	Low Medium	Low Medium	High Medium	High
Positive Small	High	Low Medium	Low	Low	Low	Low Medium	High
Zero	High	Low Medium	Low	Low	Low	Low Medium	High
Negative Small	High	Low Medium	Low	Low	Low	Low Medium	High
Negative	High	High Medium	Low Medium	Low Medium	Low Medium	Low Medium	High
Negative Big	High	High	High	High	High	High Medium	High
Negative Big	Negative	Negative Small	Zero	Positive Small	Positive	Positive Big	

Fig. 6: Rulebase for PD/PI Weighting

Finally, for the defuzzification sub-block of figure 1, we use the popular centroidal method. Defuzzification is the process of mapping the fuzzy outputs of the decision logic process from fuzzy sets to the corresponding deterministic numerical data or numerical value.

IV. SIMULATION RESULTS

Evaluation was based on performance for various speed settings and for stationary and non-stationary targets. Performance was judged on whether there was target impact or not, time to impact, and root-mean-squared (RMS) distance error to kill point at impact. For all simulation runs, we specify a target model with the physical dimensions of a typical tank (i.e., a T-80 tank), that is, a length of 24 feet, a width of 12 feet, and a height of 8 feet. The target is assumed stationary, except

for the moving target runs. The kill point or the desired impact point on the target is defined to be the geometric center of a right rectangular prism that defines the target. For all runs, the target is initially at a range of 3.5 miles at a bearing of 25 degrees. Initial missile altitude is 650 feet. The speed and altitude settings for the platform are indicative of the flight envelope of an Army surface-to-surface missile system.

A. Performance for Various Speed Command Settings:

Figure 7 shows the results for various speed command settings (i.e., V_c). Shown are bar charts of impact times and RMS distance errors at impact for the conventional guidance, fuzzy guidance with no speed control, and fuzzy guidance with fuzzy speed control. With the fuzzy speed controller, V_c is modified at each simulation tick based on current range to target and heading error, whereas V_c is held constant for the two other schemes. For all runs, all schemes were able to impact the target. As shown, the fuzzy guidance with and without the speed control achieves lower RMS impact errors compared to the conventional guidance. The discrepancy is especially noticeable at the speed setting of 400 ft/s, where an almost 50% decrease in impact error is observed. For impact times, as expected, no differences exist between the conventional and the fuzzy guidance with no speed control schemes. The fuzzy guidance with fuzzy speed control, however, attains lower impact times for speed settings of 300 and 400 ft/s. The difference in impact time is substantial (about 30%) at 300 ft/s with similar impact error performance to the conventional scheme. Impact time improvement reduces to 10% at 400 ft/s, but with a much lower than conventional scheme impact error. At the high speed setting (500 ft/s), the fuzzy guidance with speed control has a slightly higher impact time, albeit, again with a lower RMS impact error. Thus, the adaptability of the fuzzy guidance with the enhancement of the fuzzy speed controller, provides for higher accuracy with faster response time.

To gather further insight into the specifics of the fuzzy guidance and fuzzy speed controller, we plot in figure 8 simulation profiles for missile speed, PD/PI weighting factor, and gain scheduling factor from the 400 ft/s fuzzy guidance with speed control simulation run. The scheduling and weighting factors are from the fuzzy guidance heading (ψ) channel. Comparable results are also obtainable from the flight path angle (γ) channel. From the speed profile in figure 8, we see an initial dip in speed due to large initial heading errors. As these errors decrease, speed increases to approximately the maximum value of 500 ft/s. For the PD/PI weighting

factor, a value of 1.0 corresponds to total PD control, while 0.0 corresponds to total PI control. As shown, the PD/PI profile is initially high, or mostly PD control, due to large errors. As errors decrease, PI control becomes dominant, except just prior to impact where a final adjustment is required. The gain scheduling factor provides a multiplicative term to increase guidance commands based on errors and range to target. Here, we see an initial increase in gain, followed by lower gain level, and finally a higher level again just before impact. These variations correspond, in turn, to high initial errors, low error level or steady state, and final correction close to impact.

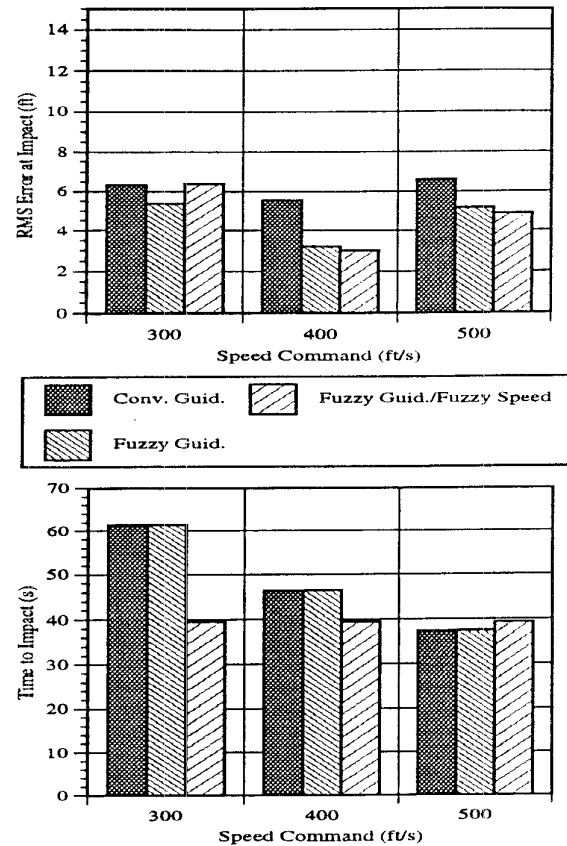


Fig. 7: Performance Evaluation - Stationary Targets

B. Performance for Non-Stationary Targets

Now we investigate performance of the the guidance schemes for moving targets. For this evaluation, we restrict ourselves to the conventional and the fuzzy guidance with fuzzy speed control schemes. We consider a scenario in which the target has an initial velocity of 50 mph with a sinusoidal acceleration of magnitude 3 ft/s². Again, we investigate impact error and time to impact performance. The resulting performance is shown in figure 9. As shown, for speed commands of 300 and 400 ft/s, the conventional guidance is unable to

impact the target. However, the fuzzy guidance with speed control achieves target impact across all speed settings. Impact times remain consistent for the fuzzy scheme across speed settings and is only slightly higher than the conventional at 500 ft/s albeit with a corresponding slightly lower impact error.

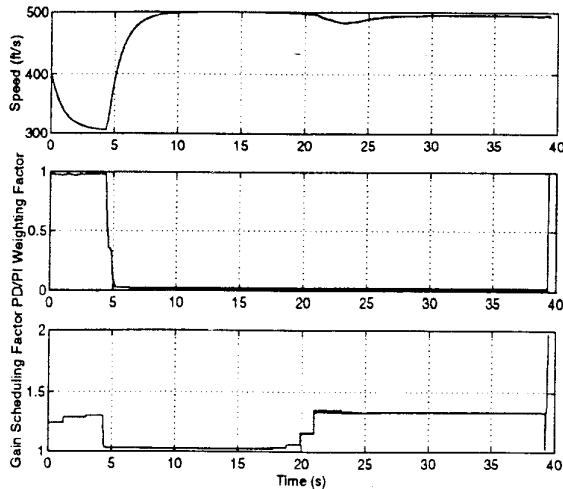


Fig. 8: Simulation Profiles

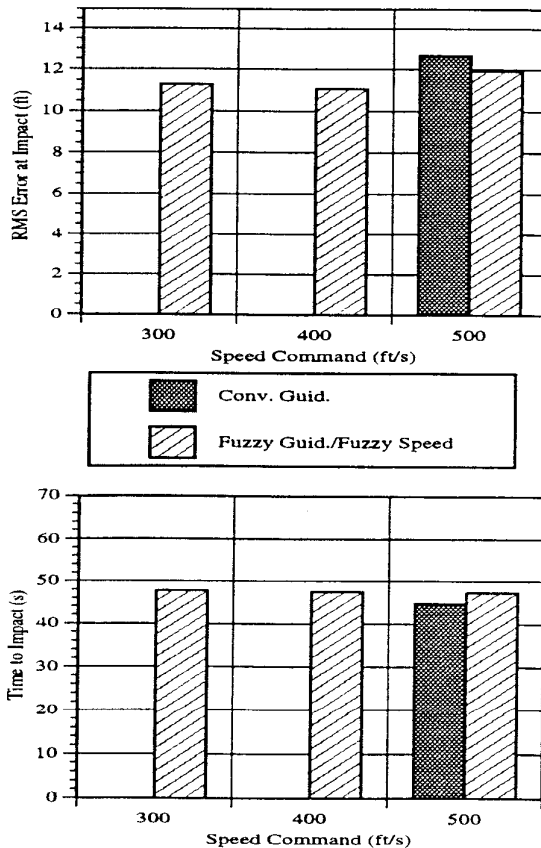


Fig. 9: Performance Evaluation - Non-Stationary Targets

V. SUMMARY AND CONCLUSIONS

Performance evaluations of the fuzzy terminal guidance scheme demonstrated the feasibility of the approach. The fuzzy guidance scheme performed at higher levels than that attained by a conventional guidance approach and demonstrated better performance as relates to desired target impact point distance error. The addition of a fuzzy speed controller provided increased performance via reduction in time to impact. Performance was demonstrated for both stationary and non-stationary ground targets. In general, this demonstration showed that a fuzzy terminal guidance system can readily achieve performance levels that meet or exceed conventional guidance approaches with the concomitant advantages such as intuitive specification of control and guidance logic, capabilities for rapid prototyping via modification of fuzzy rulebases, and robustness to sensor noise and failure accommodation.

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